

Madhushree Mondal

📍 Kolkata, West Bengal ✉ shreemadhumondal@gmail.com ☎ +91 6297193798 in LinkedIn 📄 GitHub 📁 Portfolio

About

I'm a B.Tech student in Computer Science and Engineering (IoT, Cyber Security Blockchain Technology) with interests in Full-stack development, AI/ML, Computer Vision, Robotics, Blockchain, and IoT. Passionate about building intelligent and secure systems, I have worked on projects involving human tracking, autonomous robotics, and privacy-preserving blockchain applications, with a focus on solving real-world problems through technology.

Education

University of Engineering & Management, Kolkata

July 2024 – July 2028

Bachelor of Technology in CSE (IoT, CS & Blockchain Technology); CGPA: 8.7

SGPA: 1st Sem: 8.74 — 2nd Sem: 8.74 — 3rd Sem: 8.81

Projects

Blockchain-Based Supply Chain Transparency for Agricultural Produce 🌐 Website

2025

- Developed a blockchain-based agricultural supply chain platform for secure and transparent product tracking.
- Implemented Solidity smart contracts for immutable transaction records and reduced data manipulation risks.
- Built React and Node.js modules to improve stakeholder visibility and traceability.
- Tech Stack:** Solidity, Hardhat, React, Node.js, PostgreSQL, Ethers.js — [📄 GitHub](#) [🔗](#)

Privacy Layer Using Zero-Knowledge Proofs for Multi-Chain Agricultural Blockchains

2026

- Developed a privacy-preserving blockchain application using Zero-Knowledge Proof concepts.
- Enabled confidential compliance verification across multiple blockchain networks.
- Integrated SnarkJS and Circom for improved security, tamper resistance, and data privacy.
- Tech Stack:** Solidity, Hardhat, SnarkJS, Circom, React, Node.js, Docker — [📄 GitHub](#) [🔗](#)

Human Detection, Direction Tracking & Human-Following Robot

2026

- Developed AI-based systems for real-time human detection, directional tracking, and autonomous human following.
- Integrated deep learning and computer vision techniques using OpenCV and Roboflow.
- Implemented Raspberry Pi and IoT sensor modules for intelligent tracking and navigation.
- Tech Stack:** Python, OpenCV, Roboflow, Raspberry Pi, NumPy, IoT Sensors — [📄 GitHub](#) [🔗](#)

Skills

Languages: Python, C, C++, Java, JavaScript

Core CS: Data Structures and Algorithms, Object-Oriented Programming

AI/ML: Scikit-learn, OpenCV, YOLO, Roboflow

Embedded/IoT: Arduino UNO, Raspberry Pi

Blockchain: Hardhat, Solidity, SnarkJS

Web Technologies: Node.js, HTML, CSS

Databases: MongoDB

Cloud: AWS, Google Cloud

Tools: Jupyter Notebook, Google Colab, AutoCAD, MATLAB, Proteus, ModelSim

Version Control: Git, GitHub

Achievements

- Finalist, **Smart India Hackathon 2025** – Blockchain Technology for Agriculture — [Certificate](#) [🔗](#)
- NPTEL – *The Joy of Computing using Python* (83%, Elite + Silver) — [Certificate](#) [🔗](#)

Certifications

- Fundamental Guide to AI for Professionals – Udemy (2025)
- Cyber Security Fundamentals – University of London (2025)
- Information Theory – Coursera (2026)
- Deloitte Certificate in Cyber Security (2026)